MILITARY SPECIFICATION

MOUNT, PERISCOPE: XM136

1. SCOPE

1.1 This specification covers one type of instrument known as Mount, Periscope: XM 136.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–F–13926 — Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.

MIL–P–14232 — Parts, Equipment and Tools for Ordnance Materiel, Packaging of.

MIL–I–45208 — Inspection Requirements, General Specification for.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables For Inspection by Attributes.

MIL–STD–109 — Quality Assurance Terms and Definitions.

DRAWINGS

U. S. Army Munitions Command

F8620600 — Mount, Periscope: XM 136.

LSQAP–8620600 — List of Supplementary Quality Assurance Provisions for Mount, Periscope: XM 136.

IEL–8620600 — Index of Inspection Equipment List for Mount, Periscope: XM 136.

PACKAGING DATA SHEET

MIL–P–14232/P8620600 — Packaging of Mount, Periscope: XM136

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Materials.** Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 **Fabrication.** The mount shall be manufactured in accordance with Drawing F8620-600, "Mount, Periscope: XM136", and all drawings pertaining thereto.

3.3 **General specification.** The contractor

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shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL–F–13926:

(a) Order of precedence.

(b) Dimensions and tolerances.

(c) Inorganic protective surface finishes.

(d) Part identification and marking.

(e) Workmanship.

3.4 Environmental.

3.4.1 *Operating temperatures.* The mount shall meet the applicable requirements of 3.6.1 and 3.6.2 while exposed to and thermally stabilized at ambient temperatures of —40° and +150° F.

3.4.2 *Vibration.* The mount shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5±1/4 minutes. Subsequent to vibration, there shall be no evidence of physical failure and the mount shall meet the requirements of 3.6.

3.5 Orientation. The locating surface of the mounting flange shall be secured in a horizontal position. A medium which simulates and represents the critical locating and locking surfaces on a periscope when in contact with the mount shall be utilized for insertion and removal in the mount and to determine compliance with the requirements of 3.6.1 and 3.6.2.

3.6 Performance.

3.6.1 *Periscope lock.* Rotating the periscope locking lever to the right shall lock the medium in the mount. Sliding the locking lever to the left shall position the locking lever in the final lock position.

3.6.2 *Holding features.* This requirement shall be met with the medium locked in the mount and with a 10-pound pressure applied alternately, in a horizontal, perpendicular direction, to the front and rear of the medium 7.5 inches below the lever activated locking wedges. Subsequent to the release of pressure at each point of application the medium shall be within 0.0005 inch of its original position. The measurements shall be taken adjacent to the points where pressure was applied.

3.6.3 *Shield assembly.* With the locking latch depressed (by finger pressure), the shield operating handle shall operate the shield through the full travel range (closed for protection and open for viewing). With no pressure applied on the locking trigger, the shield operating handle shall lock when the shield is in the open and closed position.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL–I–45208 and the component inspection requirements of Specification MIL–F–13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL–STD–109.

4.2 Inspection provisions.

4.2.1 *Inspection lot.* Unless otherwise spe-

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cified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" specified in Standard MIL-STD-105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926 and List of Supplementary Quality Assurance Provisions, LSQAP-8620600. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105 and the inspection level and sampling plans specified in Tables I and II titled "Classification of Defects." Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE I. *Classification of Defects*

Use Special Inspection Level S4 of Table I and Sampling Plan in Table II-A of Standard MIL-STD-105

	Requirement	Test procedure
Critical: *None*		
Major: *AQL 4.0% Defective*		
100. Operating temperature	3.4.1	4.5.1.1
Minor: *None*		

TABLE II. *Classification of Defects*

Use Inspection Level II of Table I and Sampling Plan in Table II-A of Standard MIL-STD-105

	Requirement	Test procedure
Critical: *None*		
Major: *AQL 1.0% Defective*		
101. Vibration	3.4.2	4.5.1.2
102. Periscope Lock	3.6.1	4.5.3
103. Holding Features	3.6.2	4.5.4
104. Shield Assembly	3.6.3	4.5.5
Minor: *AQL 4.0%*		
201. Paint, burrs on bearing or locating surfaces	3.3 (e)	Visual
202. Sealing compound applied to screws as per drawing	3.3 (e)	Visual

Note. The above examinations shall be conducted at standard temperatures (plus 60 degrees to 90 degrees Fahrenheit) in accordance with the requirement listed.

4.2.3 *Disposition of non-conforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL-STD-105.

4.3 **Control tests.** One Periscope Mount shall be selected at random from each 100 produced, or from each month's production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

Control test	*Requirement*	*Test procedure*
Materials	3.1	Per Drawings & Specifications
Fabrication	3.2	Visual—Per Drawings
General Specifications.	3.3	Visual—MIL-F-13926

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4.3.1 *Control test failure.* Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Inspection Equipment" requirement of Specification MIL–I–45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Government furnished inspection equipment.* Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Inspection Equipment" and "Government Furnished Material" requirements of Specification MIL–I–45208.

4.4.2 *Contractor furnished inspection equipment.*

4.4.2.1 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL–F–13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2.2 *Equipment used for environmental testing.* Testing equipment used to determine conformance with the vibration and temperature requirements shall be in accordance with the "Test Facilities" requirements outlined in Specification MIL–F–13926.

4.5 Test Methods and Procedures.

4.5.1 *Environmental.*

4.5.1.1 *Operating temperatures.* With the mount thermally stabilized at the low and high temperatures specified in 3.4.1, all movable components, locking mechanisms, and assemblies shall function without interference to determine compliance with 3.4.1.

4.5.1.2 *Vibration.* Utilizing a standard vibrating fixture, the mount shall be secured and vibrated as required by 3.4.2. Subsequent to vibration a visual and tactile inspection shall be made for physical damage and loose components or assemblies. The mount shall then meet all subsequent requirements specified herein.

4.5.2 *Orientation.* A testing device (see figure 1) shall be fabricated to support and position the mount as specified in 3.5. An adapter (simulated periscope) shall be utilized for insertion and removal with each mount to be tested. The testing device shall also provide a means of applying and measuring pressure to the adapters from opposite directions (see figure 1) as required by 3.6.2.

4.5.3 *Periscope lock.* The mount shall be positioned in the testing device (Figure 1) and secured. The locking lever shall be positioned to the left in the unlocked position. The simulated periscope (figure 2) shall be inserted upward into the mount. The locking lever will then be rotated to the right and locked by pushing or sliding the lever to the left on the elongated slot; this will lock the simulated periscope in position as specified in 3.6.1. To remove the stimulated periscope, the inspector will exert an upward pressure on the simulated periscope with one hand, slide the lever to the right and then rotate it

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to the left. The simulated periscope shall then be free for removal with both hands.

4.5.4 *Holding features.* The simulated periscope shall again be inserted into the mount as in 4.5.3. Position the dial indicator at the front or rear of the medium and set the dial to read zero (see figure 1). The dial indicator shall be graduated in increments of .0001″. Slowly apply a pressure of 10 pounds to the face opposite the dial indicator at the designated point specified in 3.6.2 and then release the pressure, read the dial indicator and record the reading. Change the dial indicator to opposite face and apply the 10 pound pressure and release. Read the error on the dial indicator. This will be total displacement error and shall not exceed the requirement specified.

4.5.5 *Shield assembly.* With the mount positioned as in 4.5.3, the inspector shall activate the shield assembly by depressing the latch with the thumb and pushing down on the handle which will raise the shield. When the shield is at full open or closed position, the latch, which is spring loaded, will recess in a position slot and lock the shield assembly as specified in 3.6.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Specification MIL–P–14232 and Packaging Data Sheet MIL–P–14232/P8620600. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The Mount, Periscope: XM136 is primarily intended for use in the fire control system of the 105mm Gun, Tank, M60 series, and is used to support the Periscope, Tank: XM39 or Periscope, Tank: XM40.

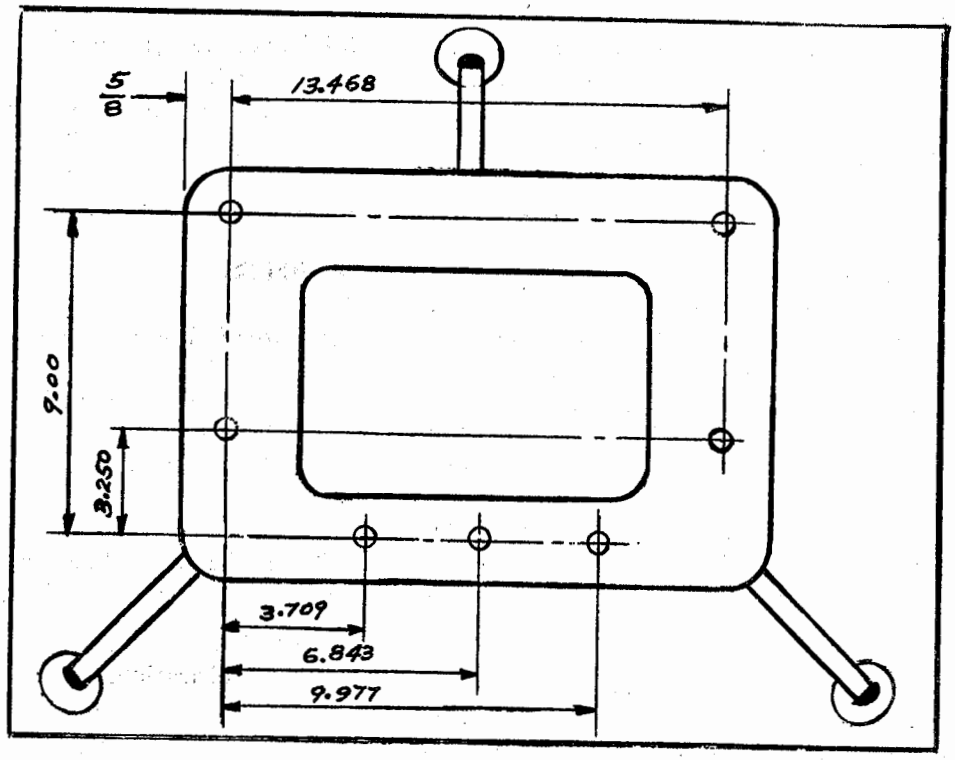
6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet number (see 5.1).

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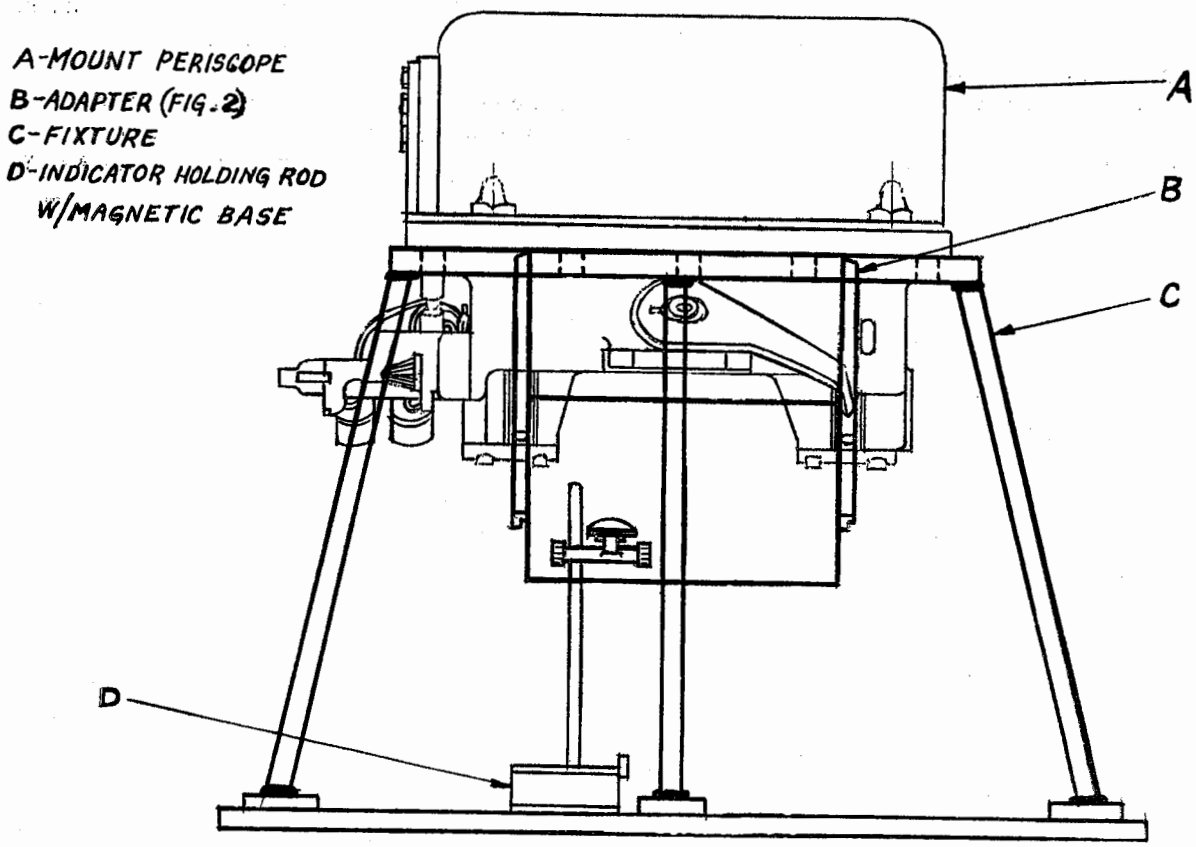


FIGURE 1

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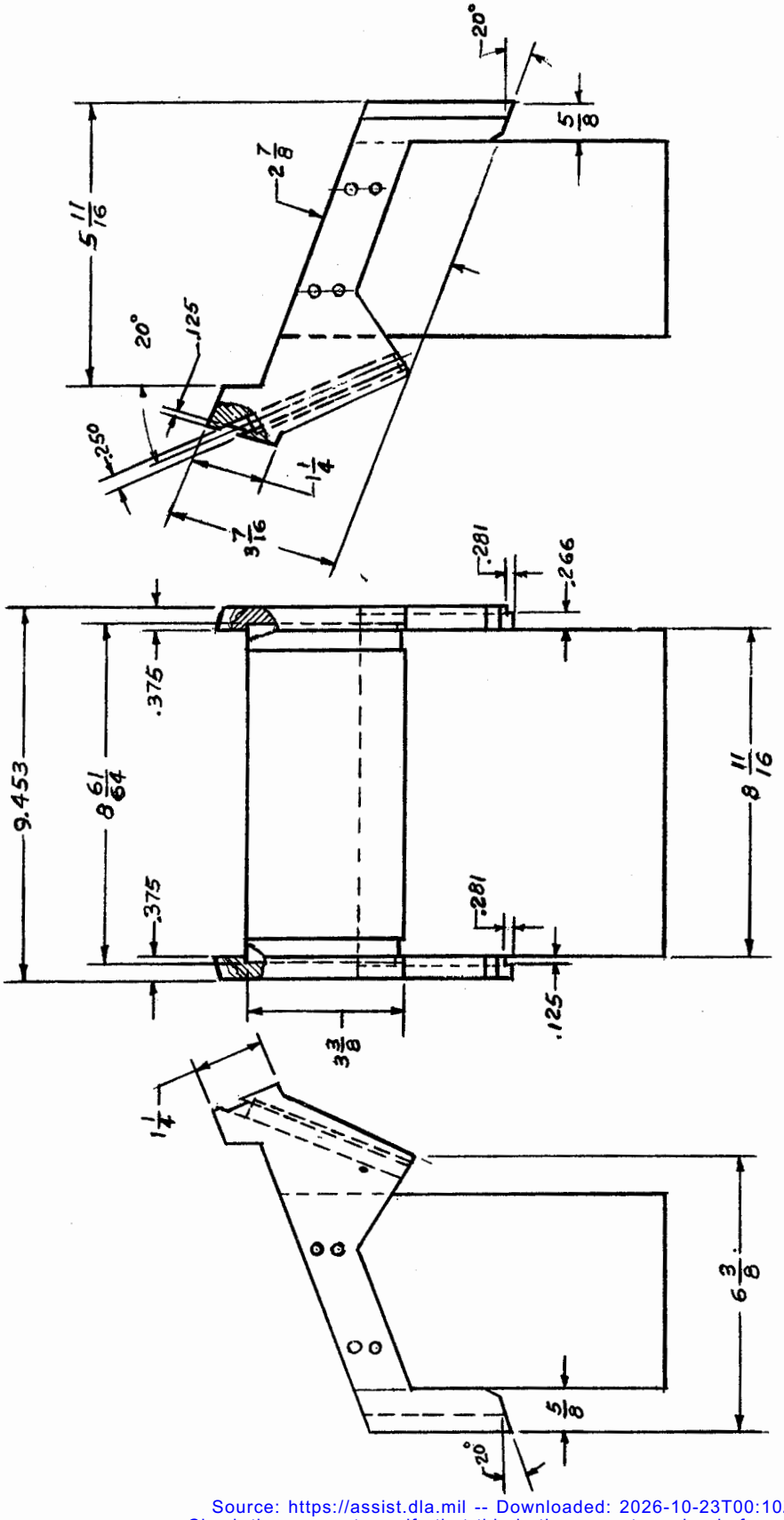


FIGURE 2.

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